

Activity 4 - NBA Database Queries

SQL Practice Questions for Midterm

Complete at least two queries from each section and submit on Canvas.

Easy - WHERE + ORDER BY

Q1 - List all active players with more than 10 seasons of experience.

Return: first_name, last_name, team_name, season_exp - ORDER BY season_exp DESC

Tables: common_player_info

```
SELECT first_name,last_name,team_name,season_exp
```

```
FROM common_player_info
```

```
WHERE rosterstatus = 'Active'
```

```
AND season_exp > 10
```

```
ORDER BY season_exp DESC;
```

Q2 - List all teams founded before the year 1960.

Return: full_name, city, year_founded - ORDER BY year_founded ASC

Tables: team

```
SELECT full_name,city,year_founded
```

```
FROM team
```

```
WHERE year_founded < 1960
```

```
ORDER BY year_founded ASC;
```

Q3 - List all players drafted in round 1 with a pick number of 5 or lower.

Return: player_name, season, overall_pick, team_name - ORDER BY season DESC, overall_pick ASC

Tables: draft_history

Q4 - List all players from the USA with more than 5 seasons of experience.

Return: first_name, last_name, position, season_exp - ORDER BY season_exp DESC

Tables: common_player_info

Easy - JOIN

Q5 - Show each player's full name alongside their draft pick number and the team that drafted them.

Return: full_name, season, overall_pick, team_name - ORDER BY overall_pick ASC

Tables: player, draft_history

```
SELECT p.full_name,d.season,d.overall_pick,d.team_name
FROM player p
JOIN draft_history d
    ON p.id = d.person_id
ORDER BY d.overall_pick ASC;
```

Q6 - Show each game with its attendance figure.

Return: team_name_home, team_name_away, pts_home, pts_away, attendance - ORDER BY attendance DESC

Tables: game, game_info

```
SELECT g.team_name_home,g.team_name_away,g.pts_home, g.pts_away, gi.attendance
FROM game g
JOIN game_info gi
    ON g.game_id = gi.game_id
ORDER BY gi.attendance DESC;
```

Q7 - Show each team's arena name and head coach.

Return: full_name, city, arena, headcoach

Tables: team, team_details

Q8 - Show each drafted player's combine results alongside their pick number.

Return: player_name, overall_pick, max_vertical_leap, bench_press - ORDER BY overall_pick ASC

Tables: draft_history, draft_combine_stats

Medium - GROUP BY + HAVING

Q9 - For each country, find total players and average seasons of experience. Only show countries with more than 10 players.

Return: country, total_players, avg_experience - ORDER BY total_players DESC

Tables: common_player_info

```
SELECT country,
COUNT(*) AS total_players,
AVG(season_exp) AS avg_experience
FROM common_player_info
GROUP BY country
```

HAVING COUNT() > 10*
ORDER BY total_players DESC;

Q10 - For each draft season, find total players drafted and average overall pick. Only show seasons where the average overall pick is greater than 30.

Return: season, total_players, avg_pick - ORDER BY season DESC

Tables: draft_history

SELECT season,
COUNT() AS total_players,*
AVG(overall_pick) AS avg_pick
FROM draft_history
GROUP BY season
HAVING AVG(overall_pick) > 30
ORDER BY season DESC;

Q11 - For each home team, find total games played and average home points. Only show teams averaging more than 110 points at home.

Return: team_name_home, total_games, avg_pts_home - ORDER BY avg_pts_home DESC

Tables: game

Q12 - For each position, find total players and average seasons of experience. Only show positions with more than 20 players.

Return: position, total_players, avg_experience - ORDER BY avg_experience DESC

Tables: common_player_info

Medium - Subqueries

Q13 - Find all players who were drafted but have no combine stats on record. (Hint: NOT EXISTS)

Return: full_name

Tables: draft_history, draft_combine_stats, player

SELECT p.full_name
FROM player p
JOIN draft_history d
ON p.id = d.person_id
WHERE NOT EXISTS (

```

SELECT 1
FROM draft_combine_stats c
WHERE c.player_id = d.person_id
);

```

Q14 - For each team, show their name and total number of home games played. (Hint: Subquery in SELECT)

Return: full_name, total_home_games

Tables: team, game

```

SELECT t.full_name,(
  SELECT COUNT(*)
  FROM game g
  WHERE g.team_id_home = t.id) AS total_home_games
FROM team t;

```

Q15 - Find teams that exist in the current system but have no historical record. (Hint: EXCEPT)

Return: nickname

Tables: team, team_history

Q16 - Find all game IDs that had both officials recorded and inactive players listed. (Hint: INTERSECT)

Return: game_id

Tables: officials, inactive_players

Q17 - Find all players whose seasons of experience is above the average for their own position. (Hint: Correlated subquery)

Return: first_name, last_name, position, season_exp

Tables: common_player_info

Q18 - For each home team, show total home wins. Only include teams with more home wins than the average home wins across all teams. (Hint: Subquery in HAVING)

Return: team_name_home, home_wins

Tables: game

Medium - Window Functions

Q19 - Rank players within each position by seasons of experience.

Return: first_name, last_name, position, season_exp, rank_in_position

Tables: *common_player_info*

```
SELECT first_name, last_name, position, season_exp,  
RANK() OVER (  
PARTITION BY position  
ORDER BY season_exp DESC  
) AS rank_in_position  
FROM common_player_info;
```

Q20 - Show each home game's points alongside the running total for that team.

Return: *team_name_home, game_date, pts_home, running_total*

Tables: *game*

```
SELECT team_name_home,  
game_date,  
pts_home,  
SUM(pts_home) OVER (  
PARTITION BY team_name_home  
ORDER BY game_date  
ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW  
) AS running_total  
FROM game;
```

Q21 - For each draft season, show each pick and the running average pick number up to that point.

Return: *season, player_name, overall_pick, running_avg_pick*

Tables: *draft_history*

Q22 - For each team, show each home game's points and the previous game's points.

Return: *team_name_home, game_date, pts_home, prev_game_pts*

Tables: *game*

Hard - CASE WHEN + GROUP BY

Q23 - For each team, show total home games per month in the 2022-23 season (season_id = '22022'). Classify monthly load as 'Heavy' (8 or more), 'Moderate' (4 - 7), or 'Light' (fewer than 4).

Return: team_name_home, month, total_games, workload - ORDER BY team_name_home, month

Tables: game

```
SELECT team_name_home,
strftime('%m', game_date) AS month,
COUNT(*) AS total_games,
CASE
  WHEN COUNT(*) >= 8 THEN 'Heavy'
  WHEN COUNT(*) BETWEEN 4 AND 7 THEN 'Moderate'
  ELSE 'Light'
END AS workload
FROM game
WHERE season_id = '22022'
GROUP BY team_name_home, strftime('%m', game_date)
ORDER BY team_name_home, month;
```

Q24 - For each draft season, categorize players by pick number as 'Lottery' (1 - 14), 'First Round' (15 - 30), or 'Second Round' (31+). Show how many players fell into each category.

Return: season, pick_category, total_players - ORDER BY season, pick_category

Tables: draft_history

```
SELECT season,
CASE
  WHEN overall_pick BETWEEN 1 AND 14 THEN 'Lottery'
  WHEN overall_pick BETWEEN 15 AND 30 THEN 'First Round'
  ELSE 'Second Round'
END AS pick_category,
COUNT(*) AS total_players
FROM draft_history
GROUP BY season,
CASE
```

```
WHEN overall_pick BETWEEN 1 AND 14 THEN 'Lottery'
WHEN overall_pick BETWEEN 15 AND 30 THEN 'First Round'
ELSE 'Second Round'
END
ORDER BY season,
CASE
WHEN pick_category = 'Lottery' THEN 1
WHEN pick_category = 'First Round' THEN 2
ELSE 3
END;
```

Q25 - For each home team, show total home wins per season and classify as 'Elite' (30 or more), 'Average' (20 - 29), or 'Bottom' (fewer than 20).

Return: team_name_home, season_id, total_wins, tier - ORDER BY season_id, total_wins
DESC

Tables: game